# Fast Scheduling and Robust Pricing: An Automated Agent Framework for Competitive Shipping Company Operations under Sealed-Bid Second-Price Auctions

Weiming Mai
Delft University of Technology
Delft, Netherlands
w.m.mai@tudelft.nl

Xueying Ou
Delft University of Technology
Delft, Netherlands
xueyingou@tudelft.nl

Ximeng Liao
Delft University of Technology
Delft, Netherlands
ximengliao@tudelft.nl

Haohui Zhang
University of Twente
Enschede, Netherlands
h.zhang-5@utwente.nl

## ABSTRACT

We present an end-to-end agent framework for the Maritime Shipping Competition (MSC) at AAMAS 2025, addressing the joint challenges of reverse second-price sealed-bid auction bidding and real-time vessel scheduling. Our approach combines a multi-start minimum-cost insertion heuristic for both pre-auction cost estimation and post-auction fleet deployment with a robust bidding strategy that integrates direct-trip costs assessments, propositional cost allocation and marginal cost pricing. We introduce a threshold-based selection mechanism to focus bids on requests that consistently appear in high-efficiency routes, and calibrate this threshold via offline simulations over varying win-rate scenarios to maximize the range of profitable outcomes.

## KEYWORDS

maritime shipping, sealed-bid second-price auctions, pickup and delivery with time windows, multi-start insertion heuristic, robust bidding, real-time scheduling

**ACM Reference Format:**
Weiming Mai, Xueying Ou, Ximeng Liao, and Haohui Zhang. 2025. Fast Scheduling and Robust Pricing: An Automated Agent Framework for Competitive Shipping Company Operations under Sealed-Bid Second-Price Auctions. In *Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), Detroit, Michigan, USA, May 19 – 23, 2025*, IFAAMAS, 4 pages.

## 1 INTRODUCTION

Shipping management is the foundation for global trade, yet faces two core challenges. First, extreme volatility in spot-market freight rates threatens profitability, which requires the development of dynamic, data-driven pricing strategies to win contracts at sustainable margins [11]. Second, assigning vessels to continuously arriving cargoes is an NP-hard problem with time-window, capacity, and precedence constraints, ruling out manual planning and calling for automated and fast heuristics [2].

The Maritime Shipping Competition (MSC) at AAMAS 2025 tasks participants with building agents that (a) bid in reverse second-price sealed-bid auctions under competitor uncertainty and (b) within a strict 60 s decision window, immediately convert awards into feasible, high-quality vessel schedules to minimize operating costs and avoid non-delivery penalties [1].

This report has two goals: (i) Devise a robust bidding strategy that guarantees profitability despite competitors' unknown bids. (ii) Propose a fast scheduling algorithm that produces high-quality vessel deployment plans in real time.

The remainder of this report is organized as follows. Section 2 reviews relevant auction and scheduling methods. Section 3 details our agent's architecture and algorithmic design.

## 2 RELATED LITERATURE

Algorithms closely related to the Maritime Shipping Competition (MSC) fall into two main categories: (i) bidding strategies; (ii) fleet scheduling algorithms.

### 2.1 Auction Strategies

The MSC employs reverse second-price sealed-bid auctions, characterized by three key properties: (i) Bidders submit bids in secret without knowing competitors' offers. (ii) The lowest bidder wins and pays the second-lowest bid. (iii) Truthful bidding is a dominant strategy in ideal settings.

*Vickrey Truth-Telling.* The classic Vickrey auction, a dominant-strategy incentive-compatible (DSIC) mechanism designed for single-item, multi-bidder environments, ensures that bidding one's true valuation is a dominant strategy under independent private costs [12]. However, accurately estimating transport costs in maritime operations is challenging. Bundling multiple cargoes on a single vessel can reduce the total cost, potentially creating a superadditive valuation compared to shipping each cargo independently. Nevertheless, allocating the resulting savings across individual bids risks losses whenever only a subset of the bundled contracts is won [3, 9].

*Heuristic and Robust Improvements.* To address the risk of losses when only part of a bundle is won, prior work uses local bundle

This work is licensed under a Creative Commons Attribution International 4.0 License.

*Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025), Y. Vorobeychik, S. Das, A. Nowé (eds.), May 19 – 23, 2025, Detroit, Michigan, USA.* © 2025 International Foundation for Autonomous Agents and Multiagent Systems (www.ifaamas.org).

heuristics that generate small cargo combinations, calculate the combined route cost, and spread the resulting savings back into individual bids [9]. Some methods refine this by scaling savings according to estimated win probabilities for each cargo [7], but such probabilities are notoriously hard to forecast due to unknown competitors strategies and potential irrational behaviors. We therefore introduce a robust bidding scheme that maintains non-negative expected profit across a wide range of plausible win probabilities, without requiring probability estimates.

## 2.2 Fleet Scheduling Methodologies

In the MSC, scheduling is split into two selective phases. *Pre-scheduling* provides cost estimates for bidding by selecting a subset of incoming cargoes and simulating tentative routes. *Post-scheduling* must then assign actually won contracts to vessels, again possibly selecting a feasible subset when not all contracts can be served. Both phases share the same underlying combinatorial structure and can be modeled as the *Selective Pickup and Delivery Problem with Time Windows and Paired Demand (SPDP-TWPD)*, where "Selective" reflects optional cargo servicing, "Pickup and Delivery" denotes the paired transport tasks, "Time Windows (TW)" capture temporal constraints, and "Paired Demand" specifies the one-to-one pickup-delivery requirement. This problem is a combinatorial optimization problem with a profit maximization goal. The MILP formulation can be found in the appendix A.

*Exact Methods.* Exact formulations for SPDP-TWPD include a compact two-index integer programming model [8] and a lexicographic bi-objective formulation balancing service rate and routing cost [5]. While these methods can find optimal solutions for small to medium instances, they are unable to deliver results within the sub-minute decision window required by MSC.

*Metaheuristic Algorithms.* To handle larger instances, researchers have developed metaheuristics such as General Variable Neighborhood Search (GVNS) [10], Ant Colony Optimization (ACO) [4], and Particle Swarm Optimization (PSO) [6]. These techniques produce high-quality schedules but still struggle to meet the real-time constraints of both pre- and post-auction scheduling in MSC.

*Greedy Insertion Heuristic.* In this work, we adopt a multi-start minimum-cost insertion heuristic that rapidly constructs multiple feasible schedules and selects the best within seconds, thus satisfying the strict timing requirements of both pre- and post-auction phases.

# 3 AGENT DECISION FRAMEWORK DESCRIPTION

We define $\mathcal{V}$ as the set of vessels, $\mathcal{R}$ as the set of shipping requests and $\mathcal{P}$ as the set of ports. At each auction round, our agent completes three tightly integrated phases—all within the strict 60-second decision window:

(1) **Pre-scheduling:** Generate a large set of route plans via multi-start minimum-cost insertion (Algorithm 3), rank them by efficiency (detailed in Section 3.1), and select the top $K$ candidates; for each selected plan $k \in [K]$, record the total cost $c_v^k$ for each vessel $v \in \mathcal{V}$ in plan $k$.

(2) **Bidding:** Choose which trades to bid on and derive individual item cost estimates by distributing each vessel's total route cost $c_v^k$ across its assigned requests inspired by propositional cost allocation and marginal cost pricing principles.

(3) **Post-scheduling:** After auction awards are revealed, assign the won requests to vessels by solving a profit-maximization scheduling problem (payment minus cost and penalty).

## 3.1 Pre-scheduling Module

This module generates $M$ candidate routing plans via multi-start minimum-cost insertion, sorts them in descending order of consolidation efficiency, and selects the top $K$. For each plan $\Pi^k$, we define

$$\text{Efficiency}(\Pi^k) = \frac{\sum_{r \in R_v^k} f^{\text{cost}}(r_i, r_j)}{\sum_{v \in \mathcal{V}} \sum_{(i,j) \in \text{leg}_v^k} f^{\text{cost}}(i,j)}, \tag{1}$$

where:

- the denominator is the total cost of executing the combined routing plan;
- the numerator sums, for each request $r$ in plan $\Pi^k$, the cost of a stand-alone trip: the vessel departs from the request's pickup node, and then directly deliver it to its delivery node.

The reason why we do not include the cost of the vehicle departing from the depot to the request's pickup node in the numerator is that in the worst case, we assume our competitors have vessels at each pickup node. An efficiency value greater than 1 indicates cost savings from bundling requests.

**Algorithm 1** `feasibility_check`

**input:** Vessel $v$, current route $\Pi_v$, assigned requests $\mathcal{R}_v$, new request $r$
**output:** best new route $\Pi'_v$, updated requests $\mathcal{R}'_v$, and cost
$best_cost \leftarrow \infty$, $best_route \leftarrow \perp$, $\mathcal{R}'_v \leftarrow \emptyset$
**for** $i = 0$ to $|\Pi_v|$ **do**
  Insert pickup $r_p$ at position $i$ in $\Pi_v$ to form $\Pi'_v$
  **for** $j = i + 1$ to $|\Pi'_v|$ **do**
    Insert delivery $r_d$ at position $j$ in $\Pi'_v$
    **if** $\Pi'_v$ satisfies TW and capacity constraints **then**
      $cost \leftarrow \text{Cost}_v(\Pi'_v)$
      **if** $cost < best_cost$ **then**
        $best_cost, best_route, \mathcal{R}'_v \leftarrow cost, \Pi'_v, \mathcal{R}_v \cup \{r\}$
      **end if**
    **end if**
  **end for**
**end for**
**return** $(best_route, \mathcal{R}'_v, best_cost)$

The multi-start minimum-cost insertion algorithm begins by taking as input a fleet of vessels, a set of transport requests (each with a pickup and delivery location, load, and time-windows), a manually defined rejection cost threshold, a function for computing the sailing and handling cost between any two nodes, and the values $M$ and $K$. It then performs $M$ independent construction runs: in each run it randomly permutes the request list and, one by one, greedily inserts

**Algorithm 2** `minimum-cost insertion`

**input:** Vessels $\mathcal{V}$, Requests $\mathcal{R}$, rejection cost $c^{\text{reject}}$, transit cost $f^{\text{cost}}(i, j)$, vessel capacity $k_v$, time windows $\text{TW}_r$, load $l_r$
Initialize $\Pi_v \leftarrow \{o_v\}$ and $\mathcal{R}_v \leftarrow \emptyset$ for all $v \in \mathcal{V}$
**for** each request $r \in \mathcal{R}$ **do**
  $best_cost \leftarrow \infty$, $best_route \leftarrow \perp$, $best_vessel \leftarrow \perp$
  **for** each vessel $v \in \mathcal{V}$ **do**
    $(\Pi'_v, cost) \leftarrow$ feasibility_check$(v, \Pi_v, \mathcal{R}_v, r)$
    **if** $cost < c^{\text{reject}}$ **and** $cost < best_cost$ **then**
      $best_cost, best_route, best_vessel \leftarrow cost, \Pi'_v, v$
    **end if**
  **end for**
  **if** $best_vessel \neq \perp$ **then**
    $\Pi_{best_vessel} \leftarrow best_route$
    $\mathcal{R}_{best_vessel} \leftarrow \mathcal{R}_{best_vessel} \cup \{r\}$
  **end if**
**end for**
**return** $\{\Pi_v \mid v \in \mathcal{V}\}$

**Algorithm 3** `multi-start minimum-cost insertion`

**input:** Vessels $\mathcal{V}$, Requests $\mathcal{R}$, manually defined reject cost $c^{\text{reject}}$, consolidation operation cost $f^{\text{cost}}(i, j)$, vessel capacity $k_v$, request time windows $\text{TW}_r$, request load $l_r$, searches $M$, selection $K$
Initialize empty list $\mathcal{RP}$ of candidate plans
**for** $m = 1$ to $M$ **do**
  $\mathcal{R}^m \leftarrow \text{Shuffle}(\mathcal{R})$
  Build plan $\Pi^m = \{\Pi_v^m\}_{v \in \mathcal{V}}$ via *minimum-cost insertion* on $\mathcal{R}^m$
  Compute per-vessel costs $\{c_v^m\}$ and add $\left(\Pi^m, \{c_v^m\}\right)$ to $\mathcal{RP}$
**end for**
Sort $\mathcal{RP}$ by overall efficiency and remove duplicates
**return** top $K$ route plans $\{\Pi^k, \{c_v^k\}\}_{k=1}^K$

each request's pickup and delivery into whichever vessel's current route yields the smallest additional cost (provided that cost stays below the rejection threshold and respects capacity and time-window constraints). Once all requests have been considered, the algorithm records the resulting route plan and computes each vessel's total cost. After completing all $M$ runs, it ranks the $M$ candidate plans by a consolidation efficiency metric and removes any duplicates. Finally, the top $K$ highest-efficiency plans—each consisting of a full set of vessel routes and their cost vectors—are returned for use in the bidding phase.

## 3.2 Bidding and Pricing Strategy

Our bidding strategy hinges on the DSIC property of Vickrey auction that we bid truthfully based on estimated cost for each request. In this section, we focus on devising a robust per-request cost estimation method.

*Threshold-based Selection and Cost Approximation.* We select for bidding only those requests whose appearance frequency $p_r$ among the $K$ pre-scheduling plans meets or exceeds the threshold $p_{\text{select}}$. For each selected request $r \in \mathcal{R}$, we compute two cost estimates:

- **Absolute cost** $\text{AbsoluteCost}_r$: the cost of a dedicated voyage from the vessel's origin through $r$'s pickup to its delivery node.
- **Marginal cost** $\text{MarginalCost}_r$: the expectation of route costs allocated to $r$ across all plans in which $r$ appears.

We calculate the route cost inspired by proportional cost allocation and marginal cost pricing principles. Specifically, all onboard requests share the delivery costs equally. However, we observe that when the vessel makes a significant detour to pick up a new request, distributing the resulting detour costs equally among all onboard requests can be unfair. To address this, we allocate the detour costs jointly between the onboard requests and the newly picked-up requests. Despite this adjustment, the absolute difference between the estimated route cost and the true cost may still be large. Therefore, we take the expected value of the route costs and blend these estimates into the final sealed-bid price to improve the robustness of our bidding strategy:

$$\widehat{c}_r = \tfrac{1}{2}\left(\text{AbsoluteCost}_r + \text{MarginalCost}_r\right),$$

thereby balancing baseline and incremental costs perspectives to approximate each request's true private cost.

**Algorithm 4** `compute_bids`

**input:** Pre-scheduling plans $\{\Pi^k\}_{k=1}^K$, Plan-appearance frequencies $\{p_r\}$, Threshold $p_{\text{select}}$, Requests $\mathcal{R}$
**output:** Bid set $\{(r, \widehat{c}_r)\}$
$\mathcal{R}_{\text{sel}} \leftarrow \{ r \in \mathcal{R} \mid p_r \geq p_{\text{select}}\}$ ▷ select by frequency
**for** each request $r \in \mathcal{R}_{\text{sel}}$ **do**
  $\text{AbsoluteCost}_r \leftarrow$ cost of vessel origin → pickup → delivery
  $\text{MarginalCost}_r \leftarrow$ average of $r$'s allocated cost over all plans where $r$ appears
  $\widehat{c}_r \leftarrow \frac{1}{2}\left(\text{AbsoluteCost}_r + \text{MarginalCost}_r\right)$
**end for**
**return** $\{(r, \widehat{c}_r) \mid r \in \mathcal{R}_{\text{sel}}\}$

*Offline Threshold Calibration.* To determine the most robust threshold $p_{\text{select}}$, we sweep candidate values in $[0, 1]$ and, for each one, simulate auction scenarios under win-rate assumptions of 10%, 20%, …, 100%:

(1) Mask each selected request with probability $1 - \rho$ to model uncertain wins.
(2) Run the auction and routing simulation to compute realized profit.
(3) Record, for each win-rate level, whether profit $\geq 0$.
(4) Determine the contiguous range of win-rate values over which profit remains non-negative, and compute its width.

The optimal $p_{\text{select}}$ is the threshold whose non-negative profit interval has the greatest width, ensuring maximal robustness across varying market conditions.

## 3.3 Post-scheduling Module

This module reuses the multi-start minimum-cost insertion framework to assign the actually won contracts to vessels. We generate $M$ candidate post-auction schedules exactly as in pre-scheduling, but

replace the "minimum cost increase" criterion with a revenue-driven insertion:

$$\Delta R(v, r) \;=\; \text{Payment}_r \;-\; \big[\text{Cost}(\Pi_v \oplus r) \;-\; \text{Cost}(\Pi_v)\big],$$

where $\Pi_v \oplus r$ denotes inserting request $r$ into vessel $v$'s current route, and $\text{Payment}_r$ is the sealed-bid price. At each insertion step we choose the vessel–request pair $(v, r)$ maximizing $\Delta R(v, r)$ (subject to feasibility).

After $M$ randomized restarts, we rank the resulting plans by total realized profit (sum of $\Delta R$ over all inserted requests) and select the top $K$. The final output is a set of $K$ high-revenue fleet schedules, each specifying the full sequence of pickups and deliveries for every vessel.

# REFERENCES


[1] [n.d.]. Maritime Shipping Competition AAMAS 2025 Guide. https://jbuerman.github.io/Maritime-Shipping-Competition-AAMAS2025/. Accessed: 2025-04-26.

[2] Krystel K Castillo-Villar, Rosa G González-Ramírez, Pablo Miranda González, and Neale R Smith. 2014. A heuristic procedure for a ship routing and scheduling problem with variable speed and discretized time windows. *Mathematical Problems in Engineering* 2014, 1 (2014), 750232.

[3] Peter Cramton, Yoav Shoham, and Richard Steinberg. 2006. *Combinatorial Auctions*. MIT Press, Cambridge, MA.

[4] M. Dorigo and L. M. Gambardella. 1997. Ant Colony System: A Cooperative Learning Approach to the Traveling Salesman Problem. *IEEE Transactions on Evolutionary Computation* 1, 1 (1997), 53–66. https://doi.org/10.1109/4235.585892

[5] R. Dumitrescu, P. Dejax, and M. Devolder. 2017. A Lexicographic Approach for the Bi-Objective Selective Pickup and Delivery Problem with Time Windows and Paired Demands. *Annals of Operations Research* 253, 1-2 (2017), 469–492. https://doi.org/10.1007/s10479-016-2253-0

[6] J. Kennedy and R. Eberhart. 1995. Particle Swarm Optimization. In *Proceedings of the IEEE International Conference on Neural Networks*. 1942–1948. https://doi.org/10.1109/ICNN.1995.488968

[7] Sébastien Lahaie and David C. Parkes. 2004. Applying Learning Algorithms to Preference Elicitation. In *Proceedings of the 5th ACM Conference on Electronic Commerce (EC '04)*. ACM, New York, NY, USA, 180–188. https://doi.org/10.1145/1007858.1007886

[8] M.-A. Manier and F. Charrier. 2018. A New Compact Two-Index Formulation for a Pickup and Delivery Problem with Time Windows and Paired Demands (SPDPTWPD). *ISTE OpenScience* 2018 (2018), 1–12.

[9] Simon Parsons, Juan A. Rodríguez-Aguilar, and Mark Klein. 2011. Auctions and Bidding: A Guide for Computer Scientists. *Comput. Surveys* 43, 2 (2011), 10:1–10:59. https://doi.org/10.1145/1922649.1922651

[10] S. Rezaee and O. Savadogo. 2016. The Multi-Vehicle Profitable Pickup and Delivery Problem. *OR Spectrum* 38, 4 (2016), 859–884. https://doi.org/10.1007/s00291-015-0411-0

[11] Wei Song, Hualong Yang, Dechang Li, and Zaili Yang. 2023. Dynamic pricing and competition of container liner shipping services in a duopoly spot market. *Computers & Industrial Engineering* 185 (2023), 109613.

[12] William Vickrey. 1961. Counterspeculation, Auctions, and Competitive Sealed Tenders. *Journal of Finance* 16, 1 (1961), 8–37.


# A PROBLEM FORMULATION

We model the problem as *Selective Pickup and Delivery Problem with Time Windows and Paired Demand (SPDP-TWPD).*

*MILP Formulation.* We define the problem as a undirected complete graph problem with $G = (N, E)$, with $N = \mathcal{P}$.

| Decision Variables | |
|---|---|
| $z_{r,v}$ | = 1, if request $r$ is assigned to vessel $v$; = 0, otherwise |
| $x^r_{i,j,v}$ | = 1, if request $r$ is shipping from port $i$ to port $j$ by vessel $v$; = 0, otherwise |
| $y_{i,j,v}$ | = 1, if vessel $v$ sails from port $i$ to port $j$; = 0, otherwise |
| $u_{i,v}$ | route order of port $i$ |
| $t^+_{i,v}$ | Departure time for vessel $v$ at port $i$ |
| $t^-_{i,v}$ | Arrival time for vessel $v$ at port $i$ |

Service time is loading / unloading time. If a port is used more than once, it needs to be duplicated with unique ID (index). If $t^{LP}_r, t^{LD}_r = None$, then we assign $t^{LP}_r, t^{LD}_r = \infty$; if $t^{EP}_r, t^{ED}_r = None$, then we assign $t^{EP}_r, t^{ED}_r = 0$ for all $r \in \mathcal{R}$.

| Parameters | |
|---|---|
| $t^{EP}_r$ | Earliest pickup time of shipping request $r$ |
| $t^{LP}_r$ | Latest pickup time of shipping request $r$ |
| $t^{ED}_r$ | Earliest delivery time of shipping request $r$ |
| $t^{LD}_r$ | Latest delivery time of shipping request $r$ |
| $t_{i,j,v}$ | Travel time of vessel $v$ from port $i$ to port $j$ |
| $t_{r,v}$ | Service time of request $r$ when using vessel $v$ |
| $l_r$ | Load of shipping request $r$ |
| $q_v$ | Capacity of vessel $v$ |
| $c_v$ | Fuel cost of $v$ with unit travel distance |
| $M$ | Big number |
| $t^{\text{last}}_v$ | Finish time of vessel $v$'s last task in the previous planning cycle expressed as a time equivalent in the current planning cycle |

$$\min \quad \sum_{v\in\mathcal{V}} c_v \cdot t^-_{d^*_v,v} + M \cdot \left(|\mathcal{R}| - \sum_{r\in\mathcal{R}}\sum_{v\in\mathcal{V}} z_{r,v}\right)$$

subject to:

$$\sum_{v\in\mathcal{V}} z_{r,v} \le 1 \qquad \forall r \in \mathcal{R}$$

$$\sum_{j\in\mathcal{P}, j\neq o_r} x^r_{o_r,j,v} = z_{r,v} \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V}$$

$$\sum_{i\in\mathcal{P}, i\neq d_r} x^r_{i,d_r,v} = z_{r,v} \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V}$$

$$\sum_{j\in\mathcal{P}, j\neq i} x^r_{i,j,v} = \sum_{j\in\mathcal{P}, j\neq i} x^r_{j,i,v} \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P}\setminus\{o_r, d_r\}$$

$$\sum_{r\in\mathcal{R}} l_r \cdot x^r_{i,j,v} \le q_v \qquad \forall v \in \mathcal{V},\ \forall i, j \in \mathcal{P},\ i \neq j$$

$$\frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} x^r_{i,j,v} \le y_{i,j,v} \qquad \forall v \in \mathcal{V},\ \forall i, j \in \mathcal{P},\ i \neq j$$

$$\sum_{j\in\mathcal{P}, j\neq o_v} y_{o_v,j,v} = 1 \qquad \forall v \in \mathcal{V}$$

$$\sum_{j\in\mathcal{P}, j\neq o_v} y_{j,o_v,v} = 0 \qquad \forall v \in \mathcal{V}$$

$$\sum_{j\in\mathcal{P}} y_{j,d_v,v} = 1 \qquad \forall v \in \mathcal{V}$$

$$\sum_{j\in\mathcal{P}} y_{d_v,j,v} = 0 \qquad \forall v \in \mathcal{V}$$

$$\sum_{j\in\mathcal{P}, j\neq i} y_{i,j,v} = \sum_{j\in\mathcal{P}, j\neq i} y_{j,i,v} \qquad \forall v \in \mathcal{V},\ \forall i \in \mathcal{P}\setminus\{o_v\}$$

$$t^-_{j,v} \ge t^+_{i,v} + t_{i,j,v} - M\cdot(1 - y_{i,j,v}) \qquad \forall v \in \mathcal{V},\ \forall i, j \in \mathcal{P},\ i \neq j,\ j \neq o_v$$

$$t^-_{i,v} \le t^{LP}_r \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = o_r$$

$$t^-_{i,v} \le t^{LD}_r \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = d_r$$

$$t^+_{i,v} \ge t^-_{i,v} \qquad \forall v \in \mathcal{V},\ \forall i \in \mathcal{P}$$

$$t^+_{i,v} \ge (t^{EP}_r + t_{r,v}) \cdot \sum_{j\in\mathcal{P}, j\neq i} x^r_{i,j,v} \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = o_r$$

$$t^+_{i,v} \ge t^-_{i,v} + t_{r,v} - M\cdot\left(1 - \sum_{j\in\mathcal{P}, j\neq i} x^r_{i,j,v}\right) \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = o_r$$

$$t^+_{i,v} \ge (t^{ED}_r + t_{r,v}) \cdot \sum_{j\in\mathcal{P}, j\neq i} x^r_{j,i,v} \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = d_r$$

$$t^+_{i,v} \ge t^-_{i,v} + t_{r,v} - M\cdot\left(1 - \sum_{j\in\mathcal{P}, j\neq i} x^r_{j,i,v}\right) \qquad \forall r \in \mathcal{R},\ \forall v \in \mathcal{V},\ \forall i \in \mathcal{P} \text{ where } i = d_r$$

$$t^+_{o_v,v} = \max\ \left\{t^{\text{last}}_v,\ 0\right\} \qquad \forall v \in \mathcal{V}$$